Explanations

Q1:

a)

```
# the 3 positive points are (1, 1), (1, 2), and (2, 1)
# the 3 negative points are (0, 0), (1, 0), and (0, 1)
# separated by diagonal gap, closest pairs are:
# (1, 1) the positive point most near negative group
# (1, 0) and (0, 1) which are the negative points most near (1, 1)
# all other points are farther out and don't touch margin
# our support vectors are the (1, 1), (1, 0), and (0, 1)
```

b)

```
# the support vectors lie on margin y * (w . x + b) = 1
# (1, 1), y = +1: w1 + w2 + b = 1
# (1, 0), y = -1: w1 + b = -1
# (0, 1), y = -1: w2 + b = -1
# by subtracting w1 + b = -1 by w2 + b = -1, we get w1=w2
# then 2w+b = 1 subtracted by w+b = -1 gives us w = 2, b = -3
```

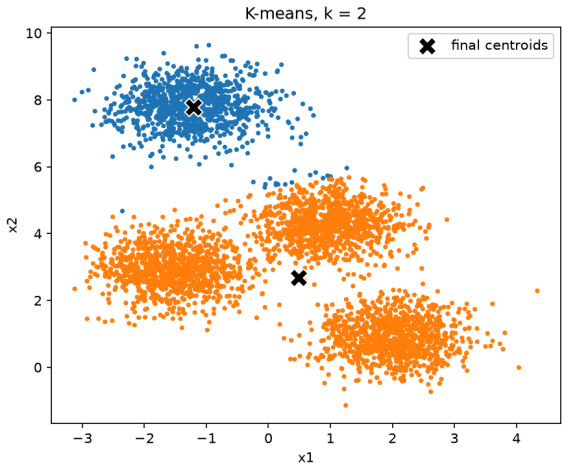
Q2:

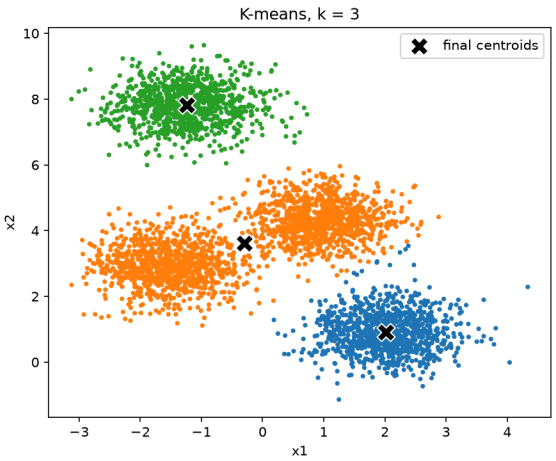
a)

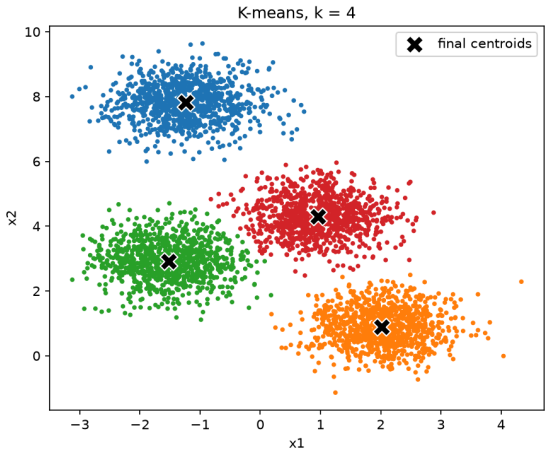
```
# for ties, if a point is equally close to multiple centroids
# it goes to the lowest index because its what np.argmin returns
# this also allows repeatable results as it is deterministic

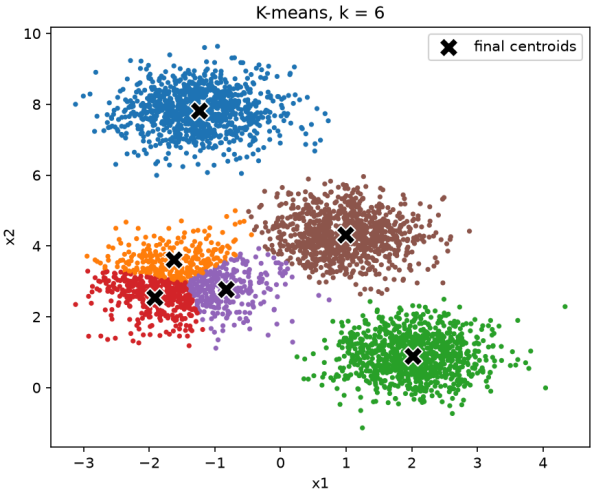
# for empty clusters, update_centroids will keep previous position
# of a centroid that received no points to avoid any issues when averaging
# no points, it also works on a copy
```

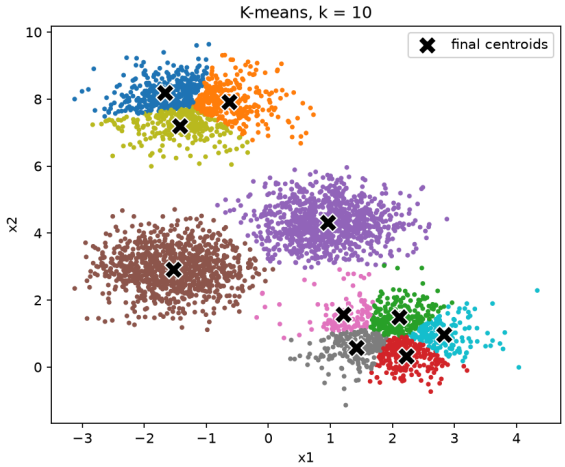
b)

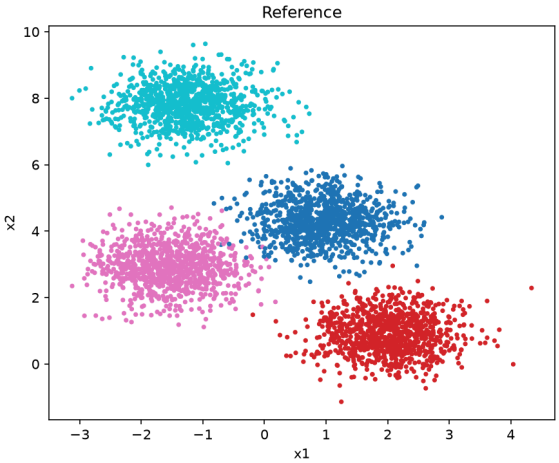













```
# as we increase the k, the clusters get smaller and centroids sit
# closer to the points, but the meaning within each cluster changes
# when k=2 the top cluster is isolated and the 3 other groups bunch
# together to one group, when k=3 the top and bottom cluster isolates
```

```
# but the middle two bunch together, at k=4 it resembles the reference
# the most matching each cluster, for k=6 and k=10 it requires a use
# of extra centroids, so it splits real groups into smaller clusters
# so while a larger k fits the data more tightly, there needs to be a
# balance and analysis as to what value of k fits the data most effectively
# low k merges distinct groups and high k fragments them
```

Q3:
The two-method comparison is the baseline against the knn repaired version. They both train the same SVM using the same features and corrupted labels, but the baseline is trained on the noisy labels whereas the knn repair replaces the noisy labels with a neighbor majority vote, then it is trained on the repaired labels.

baseline accuracy with noisy labels: 0.82795
knn repaired mean accuracy: 0.864375
difference of about 0.036

The knn repair can help because the label noise in this scenario is random, so a flipped point is still likely surrounded by neighbors that kept their labels. A majority vote of 5 neighbors can help in overruling isolated corruption.
This method can also hurt. In this case if we're working near a boundary between the classes, a point's nearest neighbor can come from both classes which makes it unpredictable. Improvement isn't guaranteed and depends on various factors like noise rate, overlapping of classes, etc. In this case, the benefit outweighed the harm giving us better accuracy.